MILESTONE 2 · DATASET PREPARATION

ContextGuard

From benchmark tables to training-ready examples

HOSPITAL · THREE ROWS

City	State	ZipCode
birmingham	al	35233
birmingham	al	35233
birmingxam	al	35233

Proposed repair: birmingham

73,979
question-and-answer examples

13
source tables: 5 core, 3 external, 5 synthetic

0
row or entity overlap between splits

1
command rebuilds everything from public sources

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github.com/SidhaarthShree07/Group-1-DS-and-AI-Lab-Project

CORE BENCHMARK TABLES

Core tables: ingestion, quality audit and ground-truth policy

5
core tables
Hospital · Flights · Beers · Rayyan · Movies

193,986
total cells
14,176 rows across 63 data columns

18,414
true errors (9.49%)
Exact match to the M1 audit

100%
deterministic
Pinned Raha commit 7be1334b8c, SHA-256 verified

TABLE	SIZE	ERRORS	ERROR RATE	WHAT THE ERRORS ARE
Hospital	1,000 × 19	**509**	2.68%	408 rule, 101 pattern; clinical measure columns
Flights	2,376 × 6	**4,920**	34.51%	Time columns only (scheduled and actual departure and arrival)
Beers	2,410 × 10	**4,362**	18.10%	Ounces format (100%), IBU missing (41.7%), ABV (28.8%)
Rayyan	1,000 × 11	**948**	8.62%	619 rule, 254 pattern, 75 missing; creation date, ISSN
Movies	7,390 × 17	**7,675**	6.11%	5,505 RatingCount commas, 856 RatingValue decimals

CLEAN-FILE GAPS

Missing clean cells: Beers 1,072 · Movies 2,680 · Rayyan 1,609 · Hospital 0 · Flights 0. Reported openly, never imputed.

RECOVERED ERRORS

1,005 Beers “N/A” and 668 Movies RatingValue errors lost in TableEG’s copies are restored from the Raha originals.

HEADERS

Columns aligned by position, not by name (Hospital 19, Movies 17, Beers 2).

Group-aware splits with zero leakage

WHY SPLIT BY ENTITY (D2)

A random row split leaks when one real entity spans many rows.

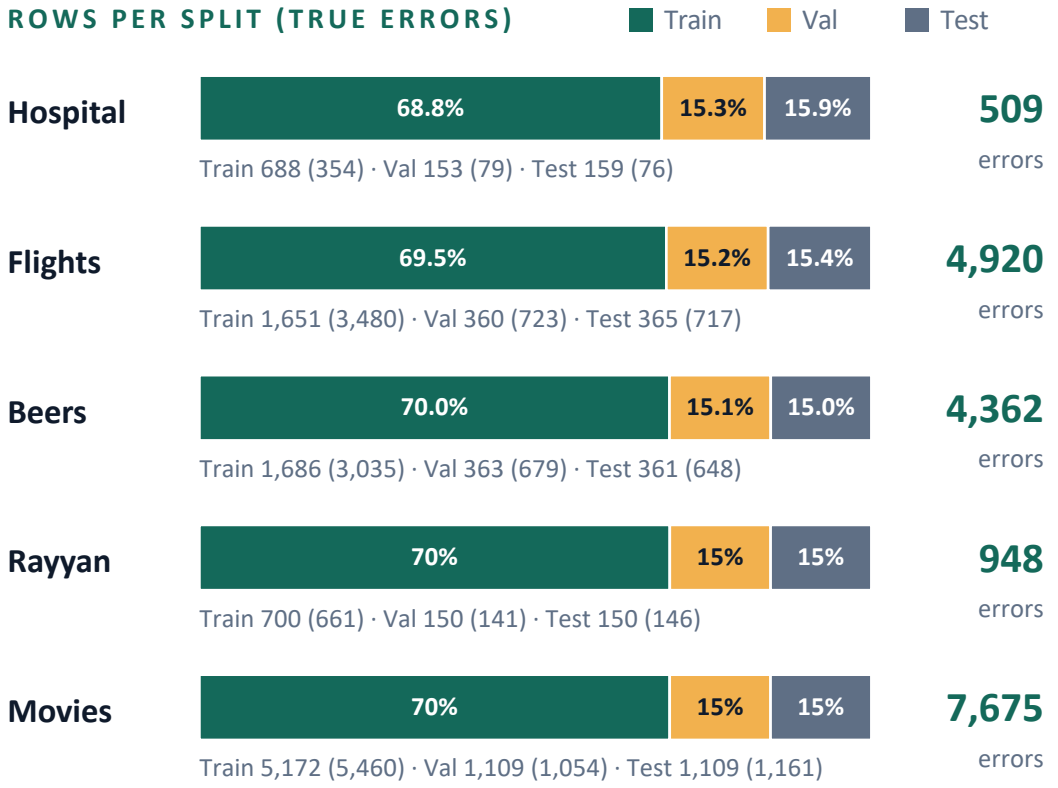
Hospital **99.9%**
provider_number · 45 entities · max 28 rows each

Flights **100%**
flight · 100 entities · max 29 rows each

Beers **94.7%**
brewery_id · 558 entities · max 62 rows each

Share of rows with an entity key used by other rows. Rayyan and Movies never repeat an id (0%), so each row is its own entity.

ROWS PER SPLIT (TRUE ERRORS)



All five tables: 18,414 errors, all conserved across splits

ZERO-LEAKAGE CHECKS

0
row overlap, all 5 tables

0
entity overlap, clean key

18,414
errors conserved in splits

29 / 29
leakage checks pass

Error rates stay close across splits, e.g. Flights 35.1 / 33.5 / 32.7%.

EXTERNAL TABLES AND LICENCES

Five more tables, each pinned and checksum-verified

TABLE	ROLE	TRUE ERRORS
Tax (20,000-row sample) 97.6% of its changes are formatting only	Training	302
Inpatient	Training	4,465
Facilities	Training	8,318
Billionaire	Unseen test	5,177
Germany (5,000 rows)	Unseen test, real errors	18,674

Splits: by clinician (Inpatient) and by facility (Facilities). Unseen tables never enter a split.

LICENCES

Raha	Apache-2.0
UCI	CC BY 4.0
Germany data	ODbL 1.0, share-alike; credit OpenStreetMap
BClean, ZeroED	No licence file; terms of the benchmarks they bundle
TableEG	Llama 3.1 licence for model outputs

Team review, 28 Oct: approved for research use and sharing with attribution

PRIVACY

Names appear in Inpatient (it looks like a public professional registry) and well-known people in Billionaire. No raw data in the repository; no personal rows in the public sample.

SYNTHETIC ERRORS

No outlier errors in the core tables, so we create them

FOUR RECIPES · CLEAN TO CORRUPTED

Missing

placeholder replaces a value

`Male` → `?`

Pattern

typo or format change

`Married-civ-spouse` → `Marred-civ-spouse`

Rule

break a found dependency

`Bachelors` → `7th-8th`

Outlier

hours_per_week out of range

`40` → `140`

THE NUMBERS

40,132 corrupted cells in 5 clean tables



Missing 12,922 · Pattern 17,076 · Rule 2,474 · Outlier 7,660

11,014 synthetic examples used, 19.99% of the 55,091 training examples (cap 20%)



USED ERRORS BY TYPE



CONTROLS AND LIMITS

- Seed 42; every record tagged source=synthetic
- adult.test never read; unseen tables never used
- Dependencies are discovered, not hard-coded (Adult 2, Bike hour 6, Wine none)
- Simple recipes are easier than real errors, so real-world results come from the unseen tables

Cheap detectors: outliers rarely help, dependencies do

OUTLIERS

Robust z-score and Isolation Forest on 11 measure columns

MOVIES RATING_VALUE · THE ONLY COLUMN WITH NUMERIC ERRORS

errors found, out of 615

Robust z-score	5 / 615
Isolation Forest	16 / 615

TAX SALARY · CORRECT VALUES FLAGGED

Isolation Forest	680 / 14,000

Most errors in numeric columns are not numbers: a % sign in Beers' ABV (such as 0.09%), placeholders in IBU.

APPROXIMATE DEPENDENCIES

g3 scan, strength at least 0.90

365 informative dependencies in the eight training tables (the first scan's 307 included 149 trivial ones with a key as determinant)



STRONG IN

Hospital · Beers · Inpatient · Facilities · Tax

ALMOST NONE IN

Rayyan · Movies

Takeaway: cheap detectors are one signal for choosing candidate cells, not a detector on their own

Fair baselines: the same held-out rows for every method

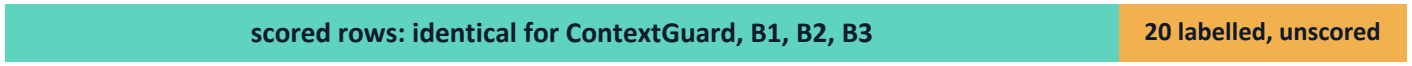
HOW COMPETITORS PREPARE DATA

Raha, Baran	Labels and scoring on the same table; 20 labelled tuples
HoloClean	Hand-written constraints, whole table
TableEG	Its released training files contain our unseen test tables, so we do not use them
Jellyfish	Hospital was in its training data
ZeroED	Zero-shot with open models; exposure of public tables to pre-training is an open risk

OUR PROTOCOL

1. All methods, ContextGuard included, are scored on each table's test split (entity-aware)
2. B3's 20 labelled rows come from the test table and are excluded from every method's scoring (Rule D8, verified for all five tables)
3. B1 gets constraints mined automatically on the test table: Hospital 67, Flights 7, Beers 9, Rayyan 0, Movies 1
4. B2 uses Raha's dBoost grid, chosen on the train split

ONE TEST SPLIT, SAME ROWS FOR EVERY METHOD (NOT TO SCALE)



Not yet done: first runs of HoloClean, dBoost and Raha on these files (Milestone 3)

FROM TABLES TO TRAINING EXAMPLES

36,795 examples from the five core tables

PIPELINE · ONE COMMAND PER STEP

1 Load and verify
`load_core.py`
Pinned sources, SHA-256, Raha comparison rule

2 Split by entity
`split.py`
70 / 15 / 15, seed 42

3 Build one record per cell
`build_examples.py`
Row, column profile, dependency evidence, similar rows

4 Check leakage
`check_leakage.py`
29 of 29 checks pass

ONE EXAMPLE (SHORTENED)

QUESTION

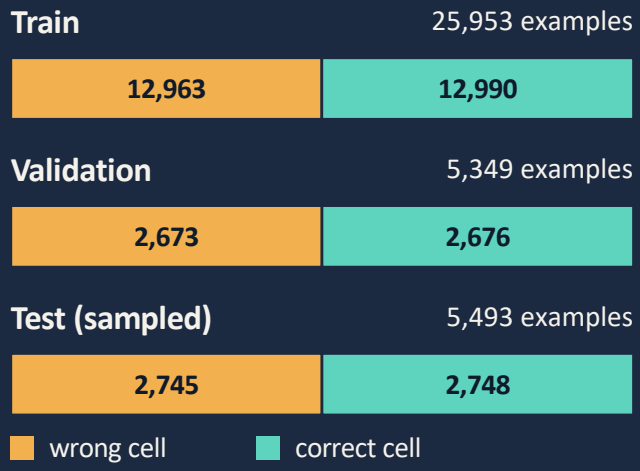
Cell `City = 'birmingxam'`

Plus its row, and the evidence: “the other rows with this ZipCode have birmingham”

ANSWER

```
{"is_error": true,
 "error_type": "rule",
 "evidence": "…",
 "repair": "birmingham"}
```

THE DATASET

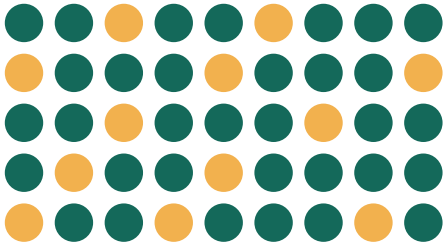


- One correct cell per wrong cell; half are hard negatives (rare but correct values)
- Prompt and answer fit the 1,500-token budget (Gemma tokens: median 484 to 759, longest 1,009)
- No clean value is ever shown to the model

A leakage check caught a real problem; all checks now pass

WHAT THE CHECK FOUND · 45 REAL HOSPITALS

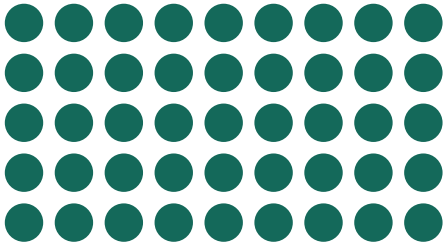
BEFORE · split on the dirty provider number



12 / 45

hospitals sat in more than one split

AFTER · split on the clean key



0 / 45

hospitals in more than one split

The dirty key has typos: 45 hospitals looked like 72. The clean key only groups rows and is never shown to the model.

THE CHECKS, ALL PASS

- ✓ Row overlap: 0
- ✓ Entity overlap (hospital, flight, brewery): 0
- ✓ Duplicate rows and prompts across splits: 0
- ✓ Unseen tables in any training file: 0
- ✓ Synthetic examples only in train: 0

DECISIONS

- Statistics for each split come from that split's own dirty table, which is what the app will see
- No outlier errors in the core tables: outliers in training must come from synthetic data
- Licences reviewed and approved; files shared as a release asset with attribution, plus a one-command rebuild

HAND-OVER CHECK

- Passed on a second machine (Aaradhya, 29 Oct): 18 of 18 tests, 55,091 training records parsed. Gemma token count: longest example 1,009 of 1,500

NEXT STEPS

Plan for Milestones 3 to 6

M3 — M4 — M5 — M6

Due 5 Nov

Model architecture

Cheap detectors D1 to D6 and candidate recall; model input format, loss and metrics; first runs of the baselines B1 to B3 on the prepared files; zero-shot test of prompt blocks (E2)

Due 19 Nov

Model training

LoRA fine-tuning runs, component experiments E1 to E3, several seeds

Due 26 Nov

Evaluation and analysis

Baseline comparison, precision–coverage curve, unseen tables, downstream experiment, error analysis

Due 10 Dec

Deployment and documentation

Gradio demo and API, model and data cards, final report

READY FOR HAND-OVER

- Hand-over check passed on a second machine (Aaradhya, 29 Oct)
- 18 of 18 tests and 29 of 29 leakage checks pass
- Longest example 1,009 Gemma tokens, limit 1,500
- A second build gives byte-identical files